

Wednesday, 30 September, 2026 at 4 pm CEST / 2 pm UTC

Registration Link: <https://washington.zoom.us/meeting/register/V768mHtxSD2P4xZelgY1Mw>

Meeting ID: 988 9348 8540 • Passcode: 953324

Find your local number: <https://us06web.zoom.us/j/kdy6pLJAeC>

SCOR Working Group 168 Webinar #11

Title: (Un)Detectability of ocean deoxygenation trends across machine learning reconstructions differing in climate predictors

Presenter:

Arianna Olivelli

Flanders Marine Institute (VLIZ)

Machine learning models have become a widely used tool for reconstructing global ocean dissolved oxygen (O₂) concentrations, yet some key methodological choices remain poorly constrained. Most existing ML-derived products include a non-physical measure of temporal changes, such as year of observation or number of months, as a model predictor, and its effect on reconstructed deoxygenation trends has not been systematically assessed. In this talk, I will present the results of a multi-model experiment comprising of eight feed-forward neural networks, which represent four different climate predictor configurations at 0.25° and 1.0° monthly resolution, and their resulting gridded products of O₂ concentrations from the surface to 1500 m depth between 1993 and 2022. I will show that while all experiments perform comparably well against a held-out test set, climate predictor choice is the dominant source of methodological uncertainty in reconstructed deoxygenation trends. Additionally, using inter-experiment consistency as a measure of signal detectability, I will show that robust detection of global ocean deoxygenation over 1993–2022 is currently limited to the surface ocean, while in the ocean interior methodological uncertainty renders robust detection of deoxygenation trends largely unachievable.

Webinar Series Information

The [4D-BGC Working Group](#) seeks to enhance access and utility of Biogeochemical (BGC) Argo observations through four-dimensional (4D) data products. These advanced data products aim to refine our understanding of ocean biogeochemistry, improve biogeochemical models and reanalysis products, and provide valuable insights for policy-making. The goal of this webinar series is to introduce new and in-development BGC data products, review techniques used to develop data products from in situ observations, and to explore ways in which 4D-BGC products are leveraged to answer scientific questions.